

SESSION 1 CLASS ACTIVITY

Leverage, WACC and Firm Value

The Situation

NOVA Components is an established manufacturer with stable operating cash flows. Management is considering a debt-financed share repurchase. The operating assets and investment policy will not change. Your group has been asked to explain what the recapitalization does to equity risk, WACC and firm value.

Part A — Perfect Capital Markets: No Corporate Taxes

Input	Value
Current debt-to-equity ratio, D/E	0.50
Cost of debt, r_D	6.0%
Current cost of equity, r_E	12.0%
Proposed debt-to-equity ratio	1.00

Use:

$$WACC = (D/V)r_D + (E/V)r_E$$

$$\text{MM Proposition II: } r_E = r_A + (D/E)(r_A - r_D)$$

1. Before calculating, predict what will happen to (i) the cost of equity, (ii) WACC and (iii) firm value when D/E rises from 0.50 to 1.00.

2. Calculate the current D/V and E/V weights and the current WACC.

3. Treat the current WACC as the required return on the operating assets, r_A . Calculate the new cost of equity at D/E = 1.00.

4. Calculate the new WACC. Does the use of more 'cheap debt' lower the firm's overall cost of capital?

5. In no more than three sentences, explain the result using the pie-model intuition from class.

Part B — Now Introduce Corporate Taxes

Keep the same operating business. Assume the corporate tax rate is 25%. For this part, use an unlevered required return $r_U = 10.0\%$ and assume debt remains at 6.0%.

Use: $r_E = r_U + (D/E)(r_U - r_D)(1 - T)$

After-tax WACC $= (D/V)r_D(1 - T) + (E/V)r_E$

For permanent debt: $V_L = V_U + TD$

6. Calculate the cost of equity at $D/E = 0.50$ and at $D/E = 1.00$.

7. Calculate after-tax WACC at both capital structures. What changes relative to Part A?

8. Suppose the unlevered firm is worth €120 million. If NOVA permanently uses €30 million of debt, calculate the value of the interest tax shield and the levered firm value.

9. Management says: “Debt lowers WACC, so NOVA should use as much debt as possible.” Based only on Session 1, what can you conclude — and what can you NOT yet conclude?

Group Deliverable — 3 Minutes

Choose one spokesperson. Present only three points: (1) what leverage does to equity risk, (2) why WACC behaves differently with and without corporate taxes, and (3) whether the tax result is enough to recommend unlimited debt.